



ODISHA SPACE APPLICATIONS CENTRE (ORSAC)

Department of Science & Technology, Govt. of Odisha

No. ORSAC/PR/1198/2024/ 1152(28) Date: 12/3/2024.

To

All the ORSAC empanelled Drone Survey Agencies (list enclosed).

Sub: For participation in the short tender relating to LiDAR Survey of Minor Irrigation Tanks.

Sir/Madam,

In inviting a reference to the subject cited above, this is to inform you that a short tender relating to LiDAR Survey of Minor Irrigation Tanks in Odisha has been published by ORSAC & enclosed for reference.

It is therefore, requested to go through the tender details and participate in the tender process within the stipulated date and time.

Yours faithfully,

SCIENTIST 'E'

12-3-24



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CONSOLIDATED LIST OF EMPANELLED VENDORS TO UNDERTAKE UAV/UAS/DRONE IMAGING SERVICES AS PER ORSAC DEFINED SOP OF GEOSPATIAL SURVEY VALID UPTO 31/01/2025

SL NO	Name Of The Vendor	Address,Phone No. and E-mail-ID
1	M/s Digital cartography and Services Pvt. Ltd.	At-1015 ,Bhagabat Sandhan,GGP Canal Road,Rasulagarh,Bhubaneswae-751025, Ph No.-9437033041, E-mail-dcsbbsr@gmail.com
2	M/s Aarav Unmanned systems Pvt. Ltd,	At-D.15/49,Dashaswamedh Ward-3,Varanasi,Uttar Pradesh-221001, Ph No.-9899855849, E-mail-info@aus.co.in
3	M/s Global Business Solutions	At-3rd floor ,Ved Arcade,#4-1-15/1,Plot no-134,Nacharam Main Road,Hydrabad,Telengana-500076, Ph No.-9966033328, E-mail-ceo@gbserp.com
4	M/s Geoid Resources Pvt. Ltd	At-Plot no-38/1918,House no-5,Binayak Enclave,Kolathia,Khandagiri,Bhubaneswar-751030, Ph No.-9437665170, E-mail- geoidresources@gmail.com
5	M/s Geokno India Pvt. Ltd.	At-Ground Floor GMR Aero Towers,RGIA,Shamshabad,Hydrabad-501218, Ph No.-7729032000, E-mail- info@geokno.com
6	M/s Skylark DronesPvt.Ltd.	At-Unit-E,Building-A,Suryakiran Apts,VIP Road,Bagulati,Kolkata-700059, Ph No.-8050058600, E-mail-info@skylarkdrones.com
7	M/s Raxa Security Services Ltd.	At-Reg. Off-25/1,Skip House ,Museum Road,Bangalore-560025, Ph No.-9845081303, E-mail-bp.mahapatra@gmrgroup.in
8	M/s Sparc Pvt. Ltd.	At-E/11,Infocity,Chandaka Industrial Estate,Chandrasekharapur,Bhubaneswar-751024, Ph No.-9437017713, E-mail- info@sparcindia.com
9	M/s Matrix Geo Solutions Pvt. Ltd	At-A1/87,3rd Floor,Sewak Park,Dwarka More,New Delhi-11059, Ph No-9599036550, E-mail- dig@matrix-geo.com
10	M/s RSI Softech India Pvt. LTd	At-Plot no-7,Type-1,IE Kukaapally,Hydrabad-500072, Ph No.-9849047844, E-mail- vinod@rsisoftech.com
11	M/s IPAGE UM SERVICES	AT-Plot No-301,Gowra Tulips,Madhapur,Hydrabad-500081, Ph No.-7799100040, E-mail- info@ipageums.com



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12	M/s Bhumi Investigation Research & Design (BIRD)	At-Plot No-1, Applolo Villa, KIIT Square, Bhubaneswar-751024, Ph No.-9438012765, E-mail- bird2017@rediffmail.com
13	Atom Aviation Services Pvt Ltd.	Plot No.-E-523, 3rd floor, Rampahalchawk, Dwarka, Sector-7, New Delhi, PIN- 110077, Ph No.-9654450079, E-mail- divya@atomaviation.com
14	My World Consultancy Services Pvt Ltd.	Telenga Pentha, Cuttack-754001
15	M/s Visiontek Consulting Services Pvt.Ltd.	Plot No.M-22/23, Chandaka Industrial Estate, Patia, Bhubaneswar-751024
16	M/s Garuda UAV Soft Solution Pvt.Ltd.	Assotech Business Cresterra, Unit No.910 - 913, 9Th floor, Tower -2, Plot no.-22, Sector-135, Noida, Uttar Pradesh-201304
17	Omni Present Robot Tech Pvt.Ltd	Plot No.-11, Behind Crown Plaza, Near IDBI Bank Basement, DDA Community Centre, Okhla Phase-1, New Delhi-110020
18	Edall Systems	Plot No-1112/584A, konena, Agrahara, Mes Colony, Bengaluru, Karnataka-560017,
19	M/s Transglobal Geomatics pvt. ltd	Hydrabad, E- Mail- info@transglobalgeomatics.com
20	M/s Dale Technologies pvt ltd	At: H.No.: 7-2-1800, Vasavi Dreams Apts. , 309 (West Block), Steet No. 1 Czech Colony, Samath Nagar, Hyderabad, 500018, Ph No. - 9247337888, E- Mail- koti@del-technologies.com
21	M/s Tricad Design Consultants pvt ltd	At: S- 307, SVSS Nivas, Main Street, Czech Colony, Opp. Gokul Theatre, Hydrabad, 500018, Ph. No- 4023817292, E-Mail- marketing@tricadinfo.com
22	M/s Pan India Consultants pvt ltd	At: 105, Udyog Nagar, Phase-VI, Gurgaon, 122015, Haryana, Ph No- 0124-2342880, E- Mail- nmspl@panindiagroup.com
23	Geomac Solutions Pvt. Ltd.	Flat No.-S-26, BDA Market Complex, Nilakantha Nagar, Nayapalli, Bhubaneswar, PIN-751012, Ph No.-0674-7960765/9438569240 Email- geomacspl@gmail.com
24	M/s Pioneer Foundation Engineers Pvt. Ltd.	Plot no.- B2, 509, 5th Boomerang, Chandivalli Farm Road, Chandivilli, Andheri(East), Mumbai, PIN-400072, Ph No.022-28581213/11, Email- sales@pfepl.com, bdpgt.promotion@gmail.com



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25	M/S Marvel Geospatial Solution Pvt. Ltd.	# 5-9-22/1/B, Shapoor Hall, Sapworwadi, Opp. New Mla Quarters, Adarsh Nagar, Hyderabad-500603, Telangana, India, Ph No-040-66413226, 23239595, Email Id- Info@Marvelgeospatial.Com
26	M/S Shanvi Resources Pvt. Ltd.	Indra Palace, 1st floor, Plot No- 1699, Kalarahanga, Bhubaneswar, Odisha, Pin-751024, Mob-0674-3104025, 9078078056, email id-info@shanviresources.com
27	M/S Geo Consultants Pvt. Ltd.	Plot No-853, Govinda Prasad, Mahavir nagar, Opposite Reliance fresh, Cuttack Puri Road, Laxmi Sagar, Bhubaneswar, 751006, Odisha, 0674-2575702, email id- consultants_geo@yahoo.com.in
28	M/s Areal Constructions and GeoIndia Services	Plot No-229, Kanan Vihar, Phase-2, Patia, Near Bank of India, Bhubaneswar- 751024, Odisha, 7008385968, 9583667618, mail id- info@acgsindia.com

ODISHA SPACE APPLICATIONS CENTRE
Deptt. of Science & Technology, Govt. of Odisha
Plot No.45/48(P), Jayadev Vihar, Near Gopabandhu Academy of
Administration, Unit-16, Bhubaneswar-751 023

SHORT TENDER NOTICE No. 05/2024

Sealed tenders are invited from eligible ORSAC empaneled UAV/Drone Survey Agencies for undertaking LiDAR Survey of Minor Irrigation Tanks under OIIPCRA through ORSAC Empaneled Agencies following Standard Operating Procedure for execution of works. The survey will be undertaken for Pre-Monsoon and Post -Monsoon period. During Pre-Monsoon, Survey is to be carried out for the **Tanks along with its Catchment area and Command area** and during Post-Monsoon, Survey is to be carried out for **the Tanks only**. The details of events are mentioned below:

- | | |
|--|------------------------------------|
| 1. Date of publication of tender notice | :12.03.2024 |
| 2. Tender paper cost | : Rs 5,000.00 |
| 3. Last date of submission of Pre-bid queries | :21.03.2024 up to 2 P.M. |
| 4. Schedule for Pre-bid meeting | :21.03.2024 at 3.30 P.M on VC mode |
| 5. Schedule for issue of Corrigendum | :22.03.2024 at 4.30 P.M. |
| 6. Last date of submission of bid documents | :29.03.2024 up to 11 A.M. |
| 7. Opening of Technical Bids | : 29.03.2024 at 11.30 A.M. |
| 8. Technical Presentation | : 29.03.2024 at 2.00 P.M |
| 9. Opening of Financial Bids of technically qualified bidders: | will be communicated later |

Non-transferable tender documents for LiDAR survey as stated above will be available from office of the Odisha Space Applications Centre (ORSAC), Bhubaneswar on payment of Rupees mentioned in the above table towards cost of tender document. The tender document cost, which is non-refundable, is payable in shape of Demand Draft drawn on any Nationalized Bank having branch at Bhubaneswar in favour of the Chief Executive, **Odisha Space Applications Centre, Bhubaneswar, Odisha**. This tender notice can also be accessed through Internet in the ORSAC website: <http://orsac.odisha.gov.in> and bid document can be downloaded. The downloaded bid document can also be used provided it is accompanied with a Demand Draft towards cost of the tender document. Tenderers interested to receive tender documents by Registered Post should pay additional amount of Rs.200/- (Rupees One hundred) only. However, ORSAC will not be responsible for postal delays in delivery of the documents.

The Tenderers are required to furnish attested copies of GST registration certificate from the competent authority as applicable along with the tender.

The deliverables as mentioned in **Annexure – A**, are to be delivered at ORSAC, Bhubaneswar.

The tenders should be submitted after due compliance with guidelines given in the tender documents. Tenders submitted in incomplete form shall be rejected outright.

The tender papers will be available for sale in the office of the ORSAC between **11.00 A.M to 4.00 P.M.** on Odisha govt. official working days' till **28.03.2024**. The sealed tenders will be received through Registered Post/Speed Post only **up to 11.00 A.M. on 29.03. 2024**. The tender will be opened at **11.30 A.M on 29.03.2024** in presence of the Tenderers or their authorized representatives, if any.

ORSAC reserves the right to cancel any or all tenders without assigning any reason thereof.

By the order of CE,


SCIENTIST 'F' 12-3-24

Terms & condition

1. Tender received by post after the stipulated date and time mentioned above will not be considered.
2. The rates quoted should be inclusive of all Taxes
3. The following documents should be furnished with the tender paper failing which the tender is liable for rejection.
 - a. Attested copy of valid ORSAC empaneled Certificate
 - b. Attested copy of valid PAN Card and GST Registration Certificates
 - c. DD/ original money receipt granted by ORSAC towards cost of tender paper.

(The original Certificates shall be produced at the time of opening of tender, if required)
4. If the date of sale, receipt and opening of the tender as specified above happens to be holiday(s), then the process of sale, receipt and opening whatsoever will be shifted to immediate next working day at the same time and venue respectively without further notice. However, date of tender paper opening may be postponed as per the convenience of the authority by intimating all the concerned tenderers through notice affixing to Notice Board / website of ORSAC.
5. The successful Bidder has to submit an PBG as mentioned in the tender document and the amount is to be forfeited if the tenderer backs out from the offer acceptance of tender by the competent authority.
6. The LiDAR survey is to be undertaken as per SOP specified by ORSAC and in the presence of User Agency, if required.
7. The tender should be submitted as per the format attached at annexure.
8. The tenderers should be file tender super scribing on the top of envelop/cover – “**Tender for LiDAR Survey**”. There shall be two bid system, one is Technical Bid and the other is financial Bid. The Technical Bid shall contain the Technical details covering sl no. 3 to 7. The financial Bid shall be as per the **Annexure – A**.
9. **BID EVALUATION-** The bids shall be evaluated based on combined **Quality and Cost Based Selection (QCBS)** method.
 - a. **Evaluation of Technical Bid** - The Bids submitted up to last date will be opened at ORSAC by the Chief Executive, ORSAC or any other officer(s) authorized by ORSAC, in the presence of such of those Bidders or their representatives who may choose to be

present at the time of opening. The representatives of the Bidder are advised to carry a letter of authority from the bidding bidders for attending the Bid opening.

The Evaluation Committee will check the submission as per the criteria. Those are selected after the check shall be called for a Technical Presentation. The committee will evaluate the technical bids as per the "**Tech Score Criteria**" mentioned in the following table.

The technical presentation will be of approximately 15 minutes' duration. Evaluation of the technical presentation shall be conducted by the evaluation committee and shall be based on Bidder's similar past experience in LiDAR Surveying, availability of infrastructure & techno-managerial capacity. ORSAC may also seek written clarifications from the Bidder soon after the bids submitted by them, during the evaluation process. The primary function or clarifications in the evaluation process is to clarify ambiguities and uncertainties, if any, arising out of the evaluation of the bid documents.

SI No	Technical Score Criteria	Max Mark	Scoring Criteria
1	The bidders must have Annual Average Turnover of Rs. 1 Crore (Rupees One Crore only) in last three financial years. (FY 2022-23, 2021-22 & 2020-21). The financial statement should reflect turnover. Auditor certified turnover statement and net worth for FY 2022-23, FY 2021-22 & FY 2020-21 in original along with balance sheet and profit and loss account for the said Financial Years to be submitted.	15	i. Up to 50 lakhs: 05 (Five) Marks ii. 50 lakhs to 01 Crore: 10 (Ten) Marks. iii. 01 Crore and Above: 15 (Fifteen) Marks.
2	The bidders should Employing Minimum 10 Numbers of DGPS/Drone Survey & Geo-informatics professionals on regular payroll, valid ESI Numbers/ Group Insurance Certificate. Copy of EPF/ESI Registration/ Group Insurance Certificate with latest return listing personnel name to be submitted	10	For DGPS/Drone survey professional: i. Up to 05 Professionals: 5 (Five) Marks ii. 06 To 10 Professionals: 8 (Eight) Marks

			iii. >10 Professionals: 10 (Ten) Marks
3	Past similar experience in LiDAR survey	25	i.0-1 Sq. Km: 10 marks, ii.1-5 Sq. Km.: 20 marks, iii. More than 5 Sq. Km: 25 marks.
4	The Organization should have at least ISO 27001:2013 certification on Information security and ISO 9001:2015 Quality Certification.	10	Producing Valid ISO Certificate= 10 Mark No Certificate=0 Mark
5	Technical Presentation, broadly covering the aspect of: • Bidder's similar past experience in Drone Survey • Proof of concept • Availability of infrastructure & techno-managerial capacity	40	It is to be awarded by Technical Evaluation Committee members and it will be the arithmetic average of all marks awarded by the Tech Committee members.
Total Technical Marks / Scores		100	

The Bidder scoring a minimum of 60 marks out of 100 in the Technical Bid Evaluation process, shall be declared as the Technically Qualified Bidder. The Financial bids shall be opened only for the Technically Qualified Bidders.

Evaluation of Price Bid: The price bids shall be opened only for the Technically Qualified Bidders on the scheduled date as mentioned in the tender notice. The authorized representatives of the Technically Qualified Bidders may be present during the price bid opening process. The price bids shall be scrutinized for their conformity to the specified formats and signatures. The price bids not in specified format and/or not with signature of the authorized representatives shall be summarily rejected. Scrutiny and evaluation of the price bids shall be conducted as follows. In the event of difference between the price mentioned in figures and words, the price in words shall be considered valid and binding.

Abnormally low quotes, 'below 80% of the estimated price' (EP-20%) shall be treated as unviable from quality and feasibility considerations and shall be summarily rejected.

If a Technically qualified bid is quoted with a price below 90% of the estimated price (i.e., EP - 10%) and in order to consider it for financial evaluation, it must accompany an additional performance security equal to the shortfall amount in form of a Bank Guarantee issued from a Nationalized bank in favour of Chief Executive, ORSAC Odisha, valid for a period of 2 years from opening of the Bid. It shall be claimed by CE, ORSAC as an unconditional forfeiture to the Govt. exchequer, if the bidder fails to execute the project as per the terms and conditions of the agreement of the execution of the work.

Thus, if the quoted price by the bidder (QP) goes below 90% of the estimated price (EP) indicated in this tender document, the value of additional security shall be = $0.9 \text{ EP} - \text{QP}$.

70% weightage will be given to Technical Score and 30% to financial response. The final score will be as per the following formula,

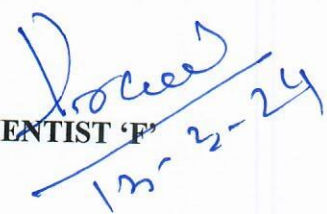
$$S = (TS \times 0.7) + (FS \times 0.3)$$

where S – Final Score, TS – Technical Score and
FS – Financial Score

- 10. Performance Bank Guarantee:** The selected Bidder shall be required to submit a Performance Bank Guarantee for an amount equal to 10% of contract value which will be valid for 3 months in excess of the working period. The Performance Bank Guarantee shall be held free of interest as security for due performance as per obligations under the contract. The Performance Bank Guarantee shall be released after successful completion of the Performance Period.
- 11. Liquidated Damages:** In case the selected Bidder fails to attain completion of the work within the scheduled time for Completion or any extension thereof due to reasons attributable to the selected bidder, ORSAC shall recover the amount of Liquidated Damages, by making deductions from the selected bidder's account or by encashment of Bank Guarantees at the rate of 0.5% of the Contract Price plus escalation, if any, excluding taxes and duties per month of delay up to a maximum of 5% of the Contract value, excluding taxes and duties. However, the payment of liquidated damages shall not in any way relieve the selected Bidder from any of its obligations to complete the Work or from any other obligations and liabilities of the Consultant under the Contract.

- 12. Insurance & Liability:** The selected Bidder shall, at his own expense, arrange appropriate comprehensive insurance to cover all risks assumed by the vendor under this contract in respect of its personnel and equipment deployed under this contract. ORSAC shall not be liable for any incident relating to any damage to the machines/ persons involved during the survey/work.
- 13. Payment Terms and Condition:** All payments shall be made by ORSAC in favor of the Bidder on successful completion of LiDAR survey and submission of relevant deliverables as per Annexure- A after QA/QC, undertaken by ORSAC for each frequency of the Survey work.
- 14.** In case of any dispute between the Tenderers and ORSAC regarding interpretation of Tender Document conditions, the decision of the Chief Executive, ORSAC shall be final and binding.
- 15.** The undersigned reserves the right to reject any or all the tenders without assigning any reason thereof. Any dispute in this matter will be settled up within Bhubaneswar jurisdiction.

By the order of CE,


SCIENTIST 'F'

Terms of Reference for conducting Li-DAR survey

S. No	Description	Specifications
1.	Benchmarks Establishment for Drone Base	From Previously Established Static DGPS base stations connected with SOI BM or as per Client's suggested Benchmarks. Required Accuracy 10mm or better. The accuracy of the ground control points (DGPS base stations & Check Points) will be checked by Network Adjustment Report/Post Processing Report submitted by service provider as deliverables (in RINEX Format).
2.	GCPs	GCP's shall be captured while flying drones over the Benchmarks established using DGPS for Corrections. 4 GCP's per Sqkm.
3.	Coverage	As per AOI with minimum of 25 m buffer all around the project area boundary demarcated with White Chuna (Limestone Powder) Line / Reflector Board
4.	GSD (Ground Sampling Distance)	2.5 cm
5.	Sensor	LiDAR & Mx Camera (RGB NIR)
6.	Image format	Drone Captured: .jpg/.jpeg/.png/ point cloud (.las) Processed Outcome: Uncompressed Geo-TIFF/ point cloud(.las)
7.	Horizontal accuracy	Two Pixels
8.	Imagery Product	Seamless mosaic covering the project area and non-overlapping, edge matched.
9.	Horizontal Datum	The World Geodetic Datum 84 (WGS-84)
10.	Map Projection	The coordinate system for all deliverables is the Universal Transverse Mercator (UTM).
11.	Vertical Datum	EGM 2008 or latest
12.	Drone Survey Outcome • Point Cloud (Classified & Colored) • Ortho Rectified Images (ORI) NCC & FCC • Digital Surface Model (DSM) • Digital Elevation Model (DEM) • Contours (Shape File) • Draping of NCC with DEM	• Raw Drone data/images along with fly log, on board GNSS/IMU data. • Point Cloud (Elevation Classified and RGB Colored) File format: .las • Post Processed for each Project area in Geo-tiff with GSD 2-5 -3.0 cm or better and Horizontal accuracy shall be 5 cm or better. • DSM & DEM with accuracy 5 cm or better • ORI in Geo-TIFF format • Contours Shape file • GeoTiff
13.	Drone Type	GNSS (RTK/PPK) Enabled Quadcopter always connected with DGPS Base Station during flight with a Dual Frequency RTK/PPK Receivers.
14.	Data Delivery Reports and Outcomes	A delivery report describing the contents of the data supplied with every data delivery (interim, staged, final).

		• Collection Report: Drone data collection report detailing mission planning and flight logs will be submitted. • Survey Report: A survey report detailing the collection of all ground control including the following will be submitted: - DGPS Base Ground Control points - Check points • Post Processing Report: - Post Processing Report detailing GNSS data, - Drone image processing, - Ortho Rectified Image generation, - DSM, DEM, Contours generation will be submitted. QA/QC Report: A QA/QC report, detailing procedures for analysis, accuracy assessment and validation of the following will be submitted. • GNSS ground control Point data (Absolute vertical accuracy/relative vertical accuracy) • Drone image processing, • Point Cloud Ortho Mosaic, DSM, DEM & Contour generation DGPS Data: • RINEX Data. • Shape Files. • RINEX Report. • Occupation sheet • Point Vector Sheet • Antenna height measurements • Instrument /antenna types & serial numbers Drone Data: • Raw Drone data • Images along with fly log, on board GNSS/IMU data
15.	Volumetric Calculation	• Report generated from the Pre and Post Drone Flight data using GIS Techniques. • Volume Calculation. • 3D Maps showing the details of the Project area with all dimensions and quantity. • DSM/DEM for Pre and Post Monsoon volumetric change • Contours for Pre and Post Monsoon volumetric change

(Full signature of Tenderer)

FORMAT OF FINANCIAL BID

Sl No.	Subject matter	Description of work	Rate quoted in Rs. (in figure & words)	Remark

The details of the enclosed bank drafts:

The details of the enclosed documents:

- 1.
- 2.
- 3.

Full name, address, phone No. of the tenderer-

(Full signature of Tenderer)